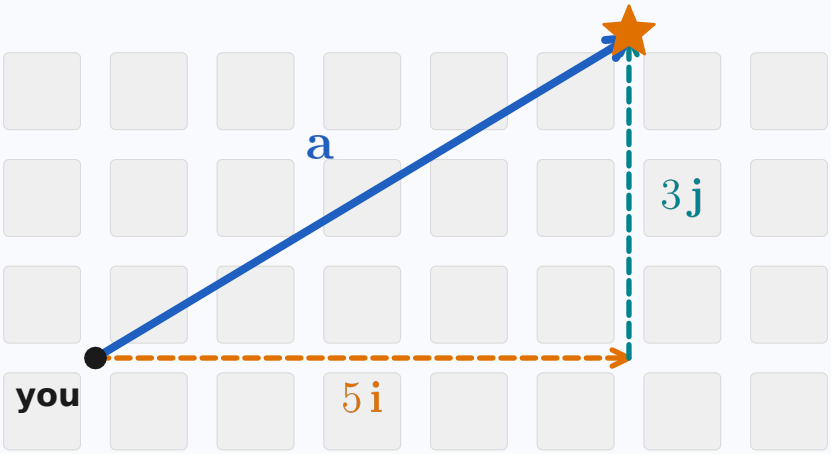
Chapter 32

Vectors

a vector is a trip: how far, and which way



5 blocks east, then 3 north.
The blue arrow is the whole trip in one go: that's the vector a = 5i + 3j.

Some things are just a size: 20 °C, 3 kg, 10 miles. Others only make sense with a **direction** too: a push, a wind, a trip across town. Those are **vectors**, and this chapter treats every one of them as a little trip on a map: how many blocks east, how many north, and (in 3D) how many floors up.

You'll add trips, stretch them, measure them, and multiply them in two very different ways (the dot product and the cross product), each with its own picture.

Goes with **Lecture 10** (scalars and vectors, notation, unit vectors, adding and subtracting, scalar multiples, dot product, cross product, determinants).

↗ Size only, or size and direction?

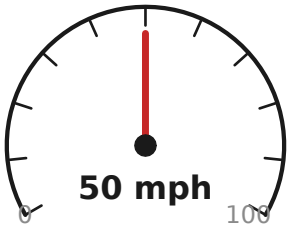
RULE: scalars and vectors

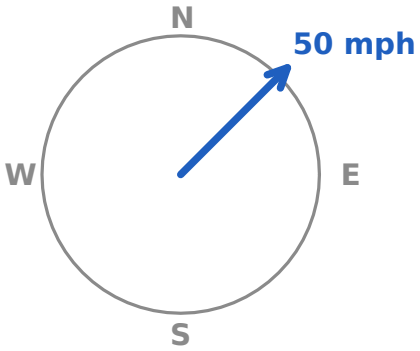
A **scalar** has size only (distance, speed, mass, temperature).

A **vector** has size **and** direction (displacement, velocity, force).

Distance is the total length travelled; **displacement** is where you end up relative to the start.

A **scalar** is one piece of information: its size. A **vector** needs two: its size **and** its direction. Same speed, two different stories:

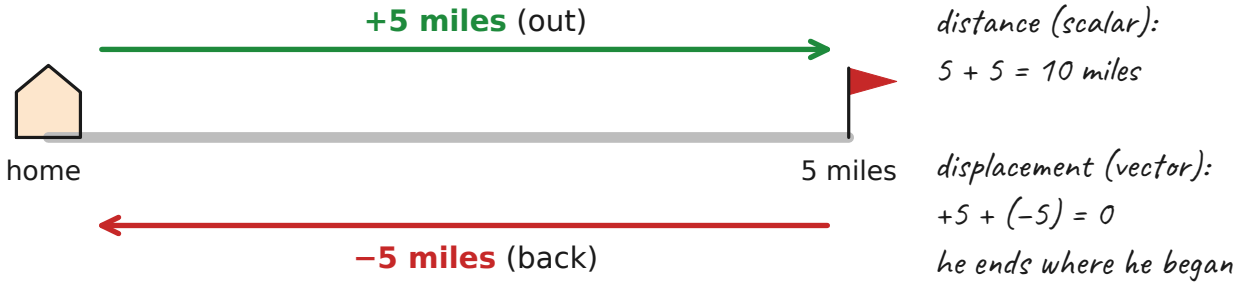




scalar: speed = 50 mph

vector: velocity = 50 mph north-east

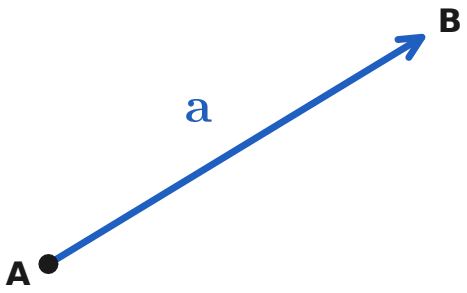
Lecture 10's runner jogs 5 miles away from home, then 5 miles back:



Same with north then south, or any direction and its opposite: the two trips cancel.

↗ One arrow, four spellings

Lecture 10 admits the notation is "quite confusing". All four of these mean the same arrow, the trip from point A to point B, named $\mathbf{a}$:



$\vec{a}$	*arrow over the letter*
$\overrightarrow{AB}$	*the trip from A to B*
$\mathbf{a}$	*bold: how books print it*
$\underline{a}$	*underlined: how you write it by hand (you can't write bold with a pen)*

↗ i, j and k: one block east, one north, one floor up

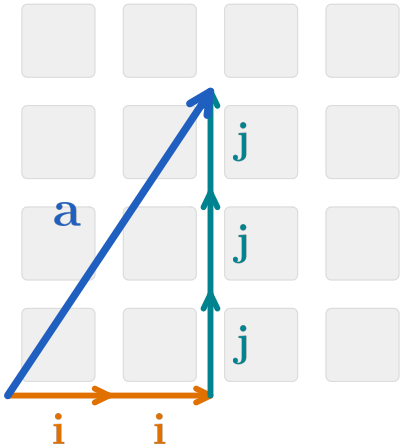
RULE: components

$\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are vectors of length 1 along the x, y and z axes.

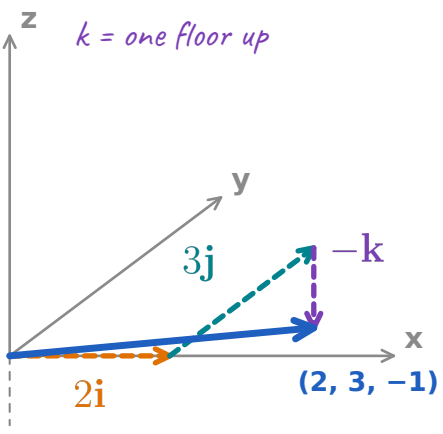
Any vector can be written $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, or as a column $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$, or as (x, y, z). The numbers are its **components**.

In a city you give directions in blocks: "2 east, 3 north". Vectors do exactly this using three standard mini-trips, each exactly **1 long** (a **unit vector**):

$\mathbf{i}$ = one step along x (east), $\mathbf{j}$ = one step along y (north), $\mathbf{k}$ = one step along z (one floor up). Lecture 10 writes them with hats: $\hat{\imath}, \hat{\jmath}, \hat{k}$.



a = 2i + 3j



On the right is Lecture 10's 3D example, $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$: 2 east, 3 north, then one floor **down** (that's the $-\mathbf{k}$). So $x = 2$, $y = 3$, $z = -1$. These are **Cartesian** coordinates; Chapter 33 shows the other way (polar).

The same trip can be written as a column (or a row), the numbers in order x, y, z:

$$2\mathbf{i} + 3\mathbf{j} - \mathbf{k} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = (2,\ 3,\ -1)$$ ← *three ways, one trip*

↗ Adding: one trip, then the next (tip to tail)

RULE: adding and subtracting vectors

Add **component by component**: $(a_x, a_y) + (b_x, b_y) = (a_x + b_x,\ a_y + b_y)$. Never mix $\mathbf{i}$ and $\mathbf{j}$ parts.

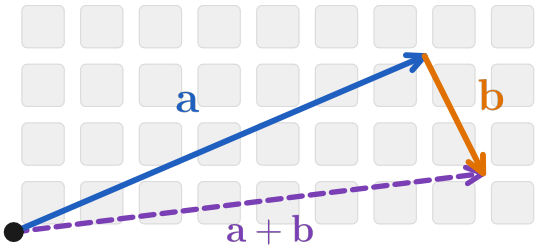
Subtract the same way: $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$, and $-\mathbf{b}$ has every sign flipped.

Order doesn't matter for adding: $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ (tip to tail either way).

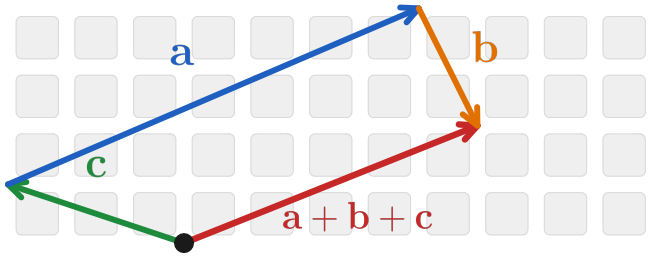
To add vectors, do the first trip, then start the second one **where the first one ended**. The answer is the single arrow from the very start to the very end. That's why "1 up + 2 right" isn't 3 of anything: it's one diagonal trip.

Lecture 10's example: $\mathbf{a} = 7\mathbf{i} + 3\mathbf{j}$, $\mathbf{b} = \mathbf{i} - 2\mathbf{j}$, $\mathbf{c} = -3\mathbf{i} + \mathbf{j}$.

a + b



c + a + b: a different order, same finish



With numbers it's even easier: keep $\mathbf{i}$'s with $\mathbf{i}$'s and $\mathbf{j}$'s with $\mathbf{j}$'s, like two separate tally columns:

add down the i column, then down the j column. Never mix them.

	$\mathbf{i}$	$\mathbf{j}$
$\mathbf{a}$	7	3
$\mathbf{b}$	1	-2
$\mathbf{c}$	-3	1
	5	2

$\mathbf{a} + \mathbf{b} + \mathbf{c} = 5\mathbf{i} + 2\mathbf{j}$

$$\mathbf{a} + \mathbf{b} = 7\mathbf{i} + 3\mathbf{j} + \mathbf{i} - 2\mathbf{j} = 8\mathbf{i} + \mathbf{j}$$ ← *i's: 7 + 1, j's: 3 − 2*

Subtracting is adding the **reverse**. $-\mathbf{c}$ is $\mathbf{c}$ turned round: same length, opposite way, so every sign flips:



$\mathbf{c} = -3\mathbf{i} + \mathbf{j}$

$-\mathbf{c} = 3\mathbf{i} - \mathbf{j}$

$$\mathbf{a} - \mathbf{c} = \mathbf{a} + (-\mathbf{c}) = 7\mathbf{i} + 3\mathbf{j} + 3\mathbf{i} - \mathbf{j} = 10\mathbf{i} + 2\mathbf{j}$$

↗ Stretching, and measuring the length

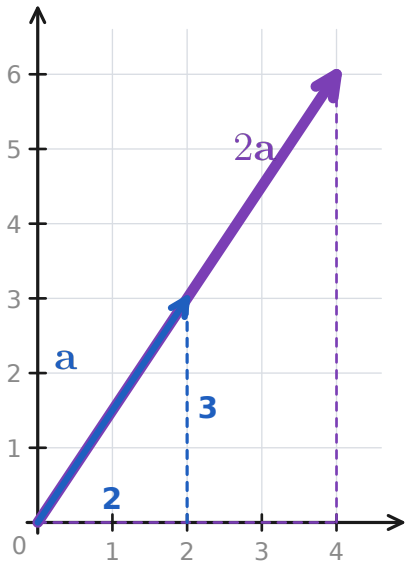
RULE: scalar multiples and magnitude

$k\mathbf{a}$: multiply **every** component by k. Same direction if $k > 0$, reversed if $k < 0$; $|k|$ times as long.

Magnitude (length): $|\mathbf{a}| = \sqrt{x^2 + y^2}$ in 2D, $\sqrt{x^2 + y^2 + z^2}$ in 3D (Pythagoras).

Multiply a vector by an ordinary number (a scalar) and you do the trip that many times: $2\mathbf{a}$ is the same direction, **twice as long**. A negative number also turns it round.

How long is the arrow? The $\mathbf{i}$ part and the $\mathbf{j}$ part meet at a right angle, so it's **Pythagoras** (Chapter 28). The length is called the **magnitude**, written $|\mathbf{a}|$.



$$|\mathbf{a}| = \sqrt{2^2 + 3^2} = \sqrt{13} \approx 3.61$$

$$|2\mathbf{a}| = \sqrt{4^2 + 6^2} = \sqrt{52} \approx 7.21$$

= 2 × 3.61: twice the trip, twice the length

$$|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$$

in 3D add the k part too (Chapter 28's box)

↗ A unit vector in any direction

RULE: the unit vector of a

The **unit vector** in the direction of $\mathbf{a}$ is $\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{|\mathbf{a}|}$: divide every component by the length.

It points the same way as $\mathbf{a}$ and has length exactly 1.

A vector of length L in the direction of $\mathbf{a}$ is $L\,\hat{\mathbf{a}}$.

Example: the unit vector in the direction of $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$.

$$|\mathbf{a}| = \sqrt{3^2 + 4^2} = 5 \quad \longleftarrow \textit{length first}$$

$$\hat{\mathbf{a}} = \tfrac{1}{5}(3\mathbf{i} + 4\mathbf{j}) = 0.6\,\mathbf{i} + 0.8\,\mathbf{j} \quad \longleftarrow \div 5$$

$$\text{check: } \sqrt{0.6^2 + 0.8^2} = \sqrt{0.36 + 0.64} = 1$$

A vector of length 10 in that direction: $10\,\hat{\mathbf{a}} = 6\mathbf{i} + 8\mathbf{j}$.

In 3D: $\mathbf{b} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

$$|\mathbf{b}| = \sqrt{4 + 1 + 4} = 3, \qquad \hat{\mathbf{b}} = \tfrac{2}{3}\mathbf{i} - \tfrac{1}{3}\mathbf{j} + \tfrac{2}{3}\mathbf{k}$$

↗ Position vectors: from point to point

RULE: position vectors, distance and midpoint

The **position vector** of a point A is the trip from the origin O to A: if A is $(2, -1)$ then $\mathbf{a} = \overrightarrow{OA} = 2\mathbf{i} - \mathbf{j}$.

The trip from A to B: $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ (**end minus start**).

The distance from A to B is its length, $|\overrightarrow{AB}| = |\mathbf{b} - \mathbf{a}|$.

The midpoint M of AB has position vector $\mathbf{m} = \dfrac{\mathbf{a} + \mathbf{b}}{2}$.

Example: A is $(2, -1)$ and B is $(5, 3)$.

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = (5-2)\mathbf{i} + (3-(-1))\mathbf{j} = 3\mathbf{i} + 4\mathbf{j}$$

end minus start

$$|\overrightarrow{AB}| = \sqrt{3^2 + 4^2} = 5$$ ← *distance AB*

$$\mathbf{m} = \tfrac{1}{2}\big((2,-1) + (5,3)\big) = \tfrac{1}{2}(7,2) = (3.5,\ 1)$$

midpoint

In 3D: A$(1,0,2)$, B$(3,-2,3)$: $\overrightarrow{AB} = (2,-2,1)$, length $\sqrt{4+4+1} = 3$, midpoint $(2,-1,2.5)$.

RULE: direction angle of a 2D vector

The angle $\mathbf{a} = x\mathbf{i} + y\mathbf{j}$ makes with the positive x-axis: $\theta = \tan^{-1}\dfrac{y}{x}$, then the **quadrant fix**.

Sketch it first. If $x < 0$ (pointing left, quadrants 2 and 3) add $180°$; if it points into quadrant 4, add $360°$ to get an angle from $0°$ to $360°$ (quadrants: Chapter 29; Chapter 33 reuses this fix for polar coordinates).

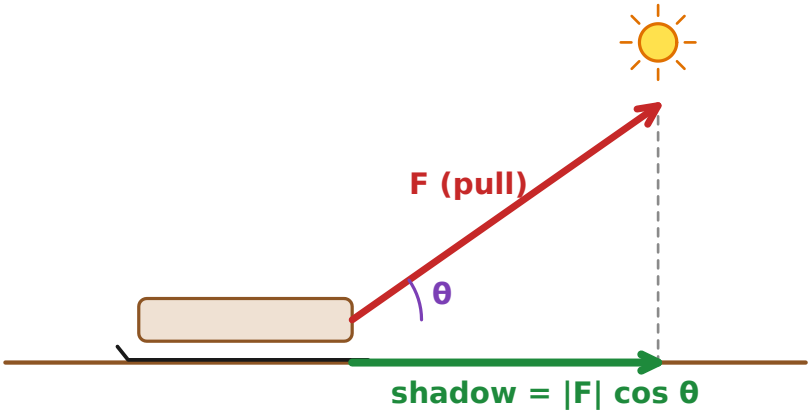
$$\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}: \quad \theta = \tan^{-1}\tfrac{4}{3} = 53.13°$$ ← *quadrant 1: as it is*

$$\mathbf{c} = -3\mathbf{i} + 4\mathbf{j}: \quad \tan^{-1}\tfrac{4}{-3} = -53.13° \to -53.13° + 180° = 126.87°$$

x < 0: add 180°

↗ The dot product: the rope's shadow

Pull a sled with a rope at an angle. Only the part of your pull **along the ground** moves the sled; the upward part is wasted. The **dot product** measures exactly that: how much one vector pulls in the direction of another.



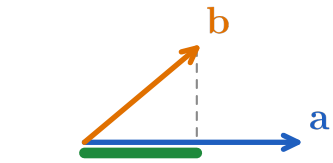
the sun is straight overhead. The rope's shadow on the ground is the useful part.

work done = $|F| \cos\theta \times$ *distance*

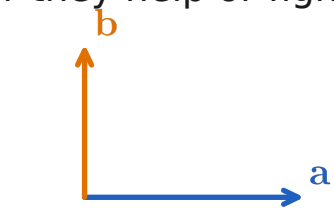
Put the two vectors tail to tail with angle θ between them. The dot product is the length of one times the length of the other's shadow on it:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$$ ← *say "a dot b"*

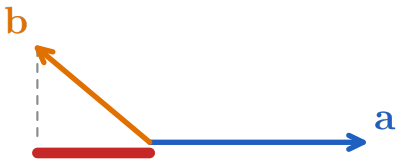
The sign tells you whether they help or fight:



positive: b helps a (angle under 90°)



zero: at right angles, no shadow at all



negative: the shadow points backwards

↗ Dot product with numbers

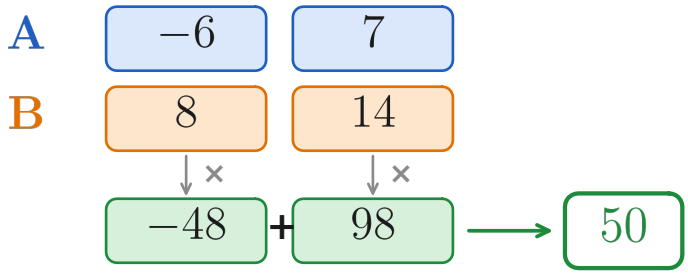
RULE: the dot product

Components: $\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y \,(+ a_z b_z)$. The answer is a **number**.

Angle form: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$, so $\cos\theta = \dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|\,|\mathbf{b}|}$.

$\mathbf{a} \cdot \mathbf{b} = 0$ means **perpendicular**; positive means the angle is under 90°; negative means over 90°.

With components you don't need the angle at all: **pair up** the matching parts, **multiply** each pair, **add**. The answer is a single number (a scalar), not an arrow.



Lecture 10: A = (−6, 7), B = (8, 14)

pair up → multiply → add = one plain number

$$\mathbf{A} \cdot \mathbf{B} = (-6)(8) + (7)(14) = -48 + 98 = 50$$

Now find the angle. Put the 50 into the angle form and undo the cosine (Chapter 28's $\cos^{-1}$):

$$|\mathbf{A}| = \sqrt{36 + 49} = 9.22, \quad |\mathbf{B}| = \sqrt{64 + 196} = 16.12$$

$$\cos\theta = \frac{50}{9.22 \times 16.12} = 0.336 \quad \leftarrow \textit{a·b ÷ both lengths}$$

$$\theta = \cos^{-1}(0.336) = 70.3°$$

Slide slip: Lecture 10 uses $\theta = 60°$ here, but $9.22 \times 16.12 \times \cos 60° = 74.3$, not 50. The true angle is 70.3°.

Lecture 10's second example: $\mathbf{A} = (2, 3)$, $\mathbf{B} = (-1, 5)$.

$$\mathbf{A} \cdot \mathbf{B} = 2 \times (-1) + 3 \times 5 = -2 + 15 = 13$$

$$\theta = \cos^{-1} \frac{13}{\sqrt{13}\sqrt{26}} = \cos^{-1} \frac{13}{3.6 \times 5.1} = \cos^{-1}(0.707) = 45°$$

The right-angle test. At 90° the shadow vanishes ($\cos 90° = 0$), so **perpendicular vectors have dot product 0**. Lecture 10: $\mathbf{a} = 2\mathbf{i} + 0\mathbf{j}$ (along x), $\mathbf{b} = 0\mathbf{i} + 1\mathbf{j}$ (along y): $2 \times 0 + 0 \times 1 = 0$.

Where it turns up: work = force · displacement (the sled), power = force · velocity, and a solar panel's energy depends on the dot product of the sun's rays with the panel's direction.

↗ Find k: perpendicular or parallel

RULE: perpendicular and parallel vectors

Perpendicular: $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$.

Parallel: $\mathbf{a} \parallel \mathbf{b} \iff \mathbf{b} = t\,\mathbf{a}$ for some number t: every component is multiplied by the **same** t (negative t: opposite directions).

PROCEDURE: finding an unknown k

1. Write both vectors with k in place.
2. Perpendicular: work out the dot product in terms of k, set it **= 0**, solve.
3. Parallel: write $\mathbf{b} = t\,\mathbf{a}$, find t from a component with no k, then use that t in the component with k.
4. Check by putting k back in.

Example 1: find k so that $3\mathbf{i} + k\mathbf{j}$ is perpendicular to $2\mathbf{i} - 6\mathbf{j}$.

$$(3)(2) + (k)(-6) = 0 \quad \longleftarrow \text{dot product = 0}$$

$$6 - 6k = 0 \;\Rightarrow\; k = 1$$

Example 2 (3D): $(k, 2, -1)$ and $(3, k, 4)$ are perpendicular.

$$3k + 2k - 4 = 0 \;\Rightarrow\; 5k = 4 \;\Rightarrow\; k = 0.8$$

Example 3: find k so that $2\mathbf{i} + k\mathbf{j}$ is parallel to $6\mathbf{i} + 9\mathbf{j}$.

$$6 = t \times 2 \;\Rightarrow\; t = 3 \quad \longleftarrow \text{x-parts give t}$$

$$9 = 3k \;\Rightarrow\; k = 3 \quad \longleftarrow \text{same t for y}$$

Example 4 (3D): $(4, -2, k)$ is parallel to $(-6, 3, 9)$. From the x-parts, $t = -6/4 = -1.5$; check y: $-2 \times (-1.5) = 3$ ✓; so $k \times (-1.5) = 9$ and $k = -6$.

✗ Common mistakes

✗ $\mathbf{a} \parallel \mathbf{b} \Rightarrow \mathbf{a} \cdot \mathbf{b} = 0$ ✓ $\mathbf{a} \parallel \mathbf{b} \Rightarrow \mathbf{b} = t\,\mathbf{a}$

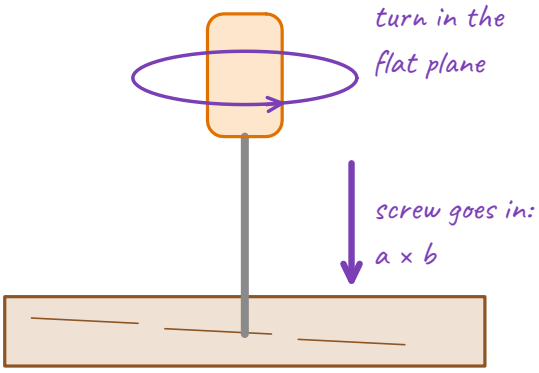
dot product 0 means perpendicular; parallel vectors have cos θ = ±1

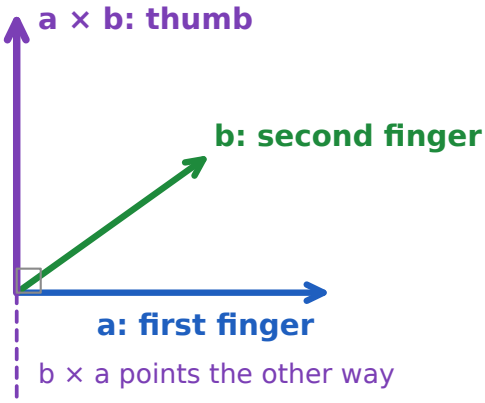
✗ $(2, k) \parallel (6, 9) :\ k = 5$ ✓ $t = 3, \quad k = 9 \div 3 = 3$

6 − 2 = 4 so k = 9 − 4? No: parallel means multiplied by the same t, not shifted

↗ The cross product: a screwdriver

The **cross product** $\mathbf{a} \times \mathbf{b}$ multiplies two vectors and gives a new **vector**, pointing at right angles to both of them. Think of a screwdriver: you turn the handle in a flat circle (that's where $\mathbf{a}$ and $\mathbf{b}$ live), and the screw moves straight in or out of the wood (that's the answer).





Right-hand rule (Lecture 10): point your right hand's first finger along $\mathbf{a}$ and your second finger along $\mathbf{b}$; your thumb points along $\mathbf{a} \times \mathbf{b}$. Swap the order and the thumb flips: $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$. (Left-handers: use your right hand anyway!)

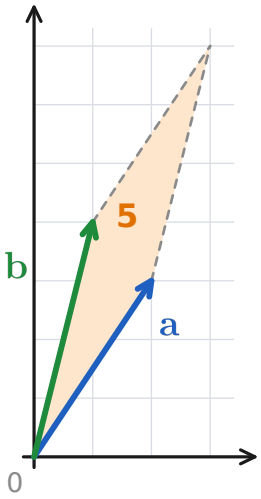
RULE: the cross product

$\mathbf{a} \times \mathbf{b}$ is a **vector** at right angles to both $\mathbf{a}$ and $\mathbf{b}$ (direction: right-hand rule). Order matters: $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$.

Size: $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}| \sin\theta$ = area of the parallelogram.

2D shortcut: for (a, b) and (c, d) the area is $|ad - bc|$ (take the size; area is never negative).

How long is it? $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}| \sin\theta$, which is exactly the **area of the parallelogram** the two arrows make. Lecture 10: $\mathbf{a} = (2, 3)$, $\mathbf{b} = (1, 4)$.



$$|\mathbf{a}|\,|\mathbf{b}| = \sqrt{13}\sqrt{17} = 14.87$$

$$\mathbf{a}\cdot\mathbf{b} = 2\times 1 + 3\times 4 = 14$$

$$\theta = \cos^{-1}\frac{14}{14.87} = 19.65^\circ$$

$$|\mathbf{a}\times\mathbf{b}| = 14.87\sin 19.65^\circ = 5.00$$

shortcut in 2D: 2×4 − 3×1 = 5 (Chapter 16's ad − bc)

Slide slip: Lecture 10's answer is 11.18, but $14.87\sin 19.7^\circ = 5.0$. The parallelogram really has area 5. Where did 11.18 come from? $14.87\times\sin(19.7) = 11.18$ with the calculator in **radians**: Chapter 27's DEG/RAD trap.

Order and sign: $(1,4)$ and $(2,3)$ give $ad - bc = 1\times 3 - 4\times 2 = -5$. The minus sign only says $\mathbf{b}$ is clockwise from $\mathbf{a}$; the area is $|-5| = 5$.

Where it turns up: turning a screwdriver or spanner (turning force in the flat plane, movement along the axis), and the force on a wire carrying current in a magnetic field.

↗ The 3 × 3 determinant: cover up and work out

In 3D, the cross product has a recipe using a 3 × 3 **determinant** (Chapter 16 did the 2 × 2 kind: $ad - bc$). Put $\mathbf{i}, \mathbf{j}, \mathbf{k}$ along the top, then the two vectors. Lecture 10: $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}$, $\mathbf{b} = 4\mathbf{i} + 9\mathbf{j} + 2\mathbf{k}$.

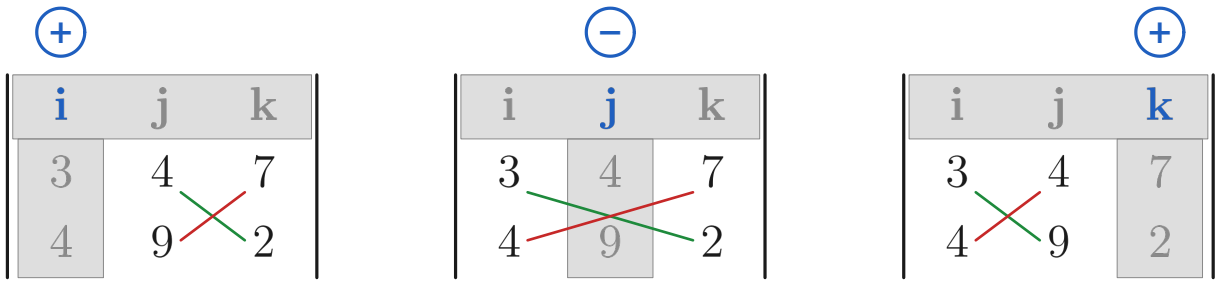
PROCEDURE: 3D cross product by determinant

1. Top row $\mathbf{i}, \mathbf{j}, \mathbf{k}$; row 2 is $\mathbf{a}$; row 3 is $\mathbf{b}$.
2. For each top-row letter, cover its row and column; work out the 2 × 2 left (**down-right minus up-right**).
3. Combine with signs **+ − +**: $+\mathbf{i}(\ldots) - \mathbf{j}(\ldots) + \mathbf{k}(\ldots)$.
4. Check: the answer dotted with $\mathbf{a}$ and with $\mathbf{b}$ must both give 0.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 4 & 7 \\ 4 & 9 & 2 \end{vmatrix}$$

← *top row: letters*
row 2: a
row 3: b

The cover-up method. For each letter on the top row, cover its row and its column with your fingers, and work out the little 2 × 2 that's left (down-right diagonal minus up-right diagonal). The signs go **+ − +**.



$+\mathbf{i}\,(4 \times 2 - 7 \times 9)$ $\quad -\mathbf{j}\,(3 \times 2 - 7 \times 4)$ $\quad +\mathbf{k}\,(3 \times 9 - 4 \times 4)$

$= -55\,\mathbf{i}$ $\quad = -(-22)\,\mathbf{j} = +22\,\mathbf{j}$ $\quad = +11\,\mathbf{k}$

$$\mathbf{a} \times \mathbf{b} = -55\,\mathbf{i} + 22\,\mathbf{j} + 11\,\mathbf{k}$$

Check it! The answer must be at right angles to **both** $\mathbf{a}$ and $\mathbf{b}$, so both dot products must be 0:

$$(-55)(3) + (22)(4) + (11)(7) = -165 + 88 + 77 = 0$$

✓ *perpendicular to a*

$$(-55)(4) + (22)(9) + (11)(2) = -220 + 198 + 22 = 0$$

✓ *perpendicular to b*

Slide slip: Lecture 10's final answer says $-11\mathbf{k}$, but the $\mathbf{k}$ part is $+(3 \times 9 - 4 \times 4) = +11$. The check catches it: with $-11\mathbf{k}$ the first dot product comes out as -154, not 0.

The general recipe, if you prefer letters:

$$\mathbf{a} \times \mathbf{b} = \mathbf{i}(a_y b_z - a_z b_y) - \mathbf{j}(a_x b_z - a_z b_x) + \mathbf{k}(a_x b_y - a_y b_x)$$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Components	$\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$
Add / scale	component by component; $k\mathbf{a}$ multiplies every part by k
Magnitude	$\lvert\mathbf{a}\rvert = \sqrt{x^2 + y^2 + z^2}$
Unit vector	$\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{\lvert\mathbf{a}\rvert}$ length L in that direction: $L\hat{\mathbf{a}}$
Position vectors	$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}, \qquad \mathbf{m} = \dfrac{\mathbf{a} + \mathbf{b}}{2}$ distance $AB = \lvert\mathbf{b} - \mathbf{a}\rvert$
Direction angle	$\theta = \tan^{-1}\dfrac{y}{x}$ then add 180° if $x < 0$, 360° if in quadrant 4
Dot product	$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \cos\theta$
Angle between	$\cos\theta = \dfrac{\mathbf{a} \cdot \mathbf{b}}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$
Perpendicular / parallel	$\mathbf{a} \cdot \mathbf{b} = 0$ / $\mathbf{b} = t\mathbf{a}$
Cross product size	$\lvert\mathbf{a} \times \mathbf{b}\rvert = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \sin\theta$ parallelogram area; 2D: $\lvert ad - bc\rvert$
3D cross product	determinant with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ on top, signs + − +; check both dot products are 0

↗ Your turn

Sketch every vector on squared paper first. Answers in grey.

Trips, adding, lengths

1. $p = 3i - 2j$ and $q = -i + 5j$. Find $p + q$ and $p - q$.
2. Find $|p|$, exactly and to 2 d.p.
3. Find $3q$ and its length.

1. $2i + 3j$; $4i - 7j$ 2. $\sqrt{13} \approx 3.61$ 3. $-3i + 15j$; $\sqrt{234} \approx 15.30$

Dot product

4. $(4, 1) \cdot (2, -3)$
5. The angle between $(1, 2)$ and $(3, 1)$
6. Are $2i + 3j$ and $6i - 4j$ perpendicular?

4. $8 - 3 = 5$ 5. $\cos^{-1}(5 \div (\sqrt{5} \times \sqrt{10})) = 45°$ 6. yes: $12 - 12 = 0$

Cross product

7. Area of the parallelogram made by $(3, 1)$ and $(1, 2)$
8. $(1, 2, 0) \times (0, 1, 3)$ by the cover-up method
9. Check your answer to 8 with two dot products.

7. $3 \times 2 - 1 \times 1 = 5$ 8. $6i - 3j + k$ 9. $6 - 6 + 0 = 0$ and $0 - 3 + 3 = 0$

Distance or displacement?

10. A bus drives 3 km east then 4 km north. Distance travelled? Size of the displacement?

10. 7 km; $\sqrt{(9 + 16)} = 5$ km

Find the unknown

11. Find k so that $ki + 4j$ is perpendicular to $2i - j$.
12. Find k so that $ki + 6j$ is parallel to $2i + 3j$.
13. Find the unit vector in the direction of $5i - 12j$.
14. A is $(1, 3)$ and B is $(7, -5)$. Find AB, the distance AB and the midpoint.
15. Find the vector of length 15 in the direction of $3i + 4j$.
16. Find the direction angle of $-2i - 2j$ ($0°$ to $360°$).

11. $2k - 4 = 0$, $k = 2$ 12. $t = 2$ (from $3 \to 6$), so $k = 4$ 13. length 13: $(5/13)i - (12/13)j$ 14. $b - a = 6i - 8j$; $\sqrt{(36 + 64)} = 10$; $((1 + 7)/2, (3 - 5)/2) = (4, -1)$ 15. $15 \times (0.6i + 0.8j) = 9i + 12j$ 16. $\tan^{-1}(1) = 45°$; $x < 0$ so $+ 180°$: $225°$